

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 17: Forests: Our Lifeline

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	17 — Forests: Our Lifeline
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. This is the hardest tier: expect quantitative energy-flow estimates, two- and three-step food-web ripples that loop back through shared predators, nutrient-cycle breaks disguised as harmless tidying, the standing confusion of energy flow (one-way, lost as heat) with nutrient flow (cyclic), and the night-respiration paradox dressed up as proof that forests harm the air — every trap is a plausible half-truth, so reason to the end before choosing. Do not write on this paper — mark answers on your answer sheet.

1. A light meter reads 100 units above the canopy, 40 units in the understory, 12 units in the shrub layer and 3 units at the herb level. A gardener wants to transplant a fern that, in greenhouse tests, photosynthesises best at about 10 to 15 units and is scorched above 50 units. The single best planting spot, and the reason, is:

- (A) above the canopy at 100 units, because more light always means more food and faster growth for any plant
- (B) the herb level at 3 units, because the darkest spot is always the safest for a shade plant
- (C) the understory at 40 units, because a value roughly midway down the gradient must suit a middle-sized plant
- (D) the shrub layer at about 12 units, because it falls inside the fern's best range while staying well below the level that scorches it

2. A forest holds roughly 60000 kg of green plants, which support deer totalling about 6000 kg, which support leopards totalling about 600 kg. A student reasons that if the plant mass could somehow be tripled, the leopard mass would also exactly triple to 1800 kg. The most accurate critique is:

- (A) the rough tenfold drop at each step would still hold, so tripling the producer base would tend to raise the leopard mass towards roughly 1800 kg, but real losses vary so it is an estimate, not an exact figure
- (B) the leopard mass cannot change at all, because the number of top predators is fixed by instinct regardless of how much food exists
- (C) tripling the plants would leave the leopards unchanged, because carnivores draw energy from prey and so are cut off from the plant supply
- (D) the leopard mass would rise far more than threefold, because energy is gained rather than lost as it moves up the chain

3. In the chain grass to grasshopper to frog to snake to hawk, the grass first captures the Sun's energy and about nine-tenths is lost as heat at each feeding step. Of the original captured energy, the share that finally reaches the hawk's body is closest to:

- (A) about half, because energy is shared roughly equally between the producer and the top animal
- (B) roughly one part in ten thousand, because four feeding steps each pass on only about a tenth
- (C) almost all of it, because the hawk sits at the top and so collects the energy from everything below it
- (D) none at all, because energy stops moving once it enters the first animal

4. A forest web: clover is eaten by both a vole and a rabbit; the vole is eaten by an owl and by a fox; the rabbit is eaten only by the fox. A disease removes all the foxes. Tracing two steps, the most defensible prediction is:

- (A) voles and rabbits rise by exactly the same amount, since both were prey of the fox and so both gain equally
- (B) the clover increases, because losing the foxes means the owl now eats far more of the plant-grazing voles
- (C) rabbits, having lost their only predator, tend to rise more sharply than voles, which still face the owl, and the heavier rabbit grazing presses harder on the clover
- (D) the owl dies out, because the fox had been the owl's main source of food

5. A teacher draws two arrows for the same forest: one for the path of energy, one for the path of mineral nutrients. A student insists both arrows must form closed loops because both pass through the same plants and animals. The precise correction is:

- (A) both energy and nutrients loop, since anything moving through a food chain must eventually come back to its start
- (B) energy cycles back to the soil while nutrients flow one way and are lost from the forest for good

- (C) neither energy nor nutrients moves at all once it has entered a green plant
- (D) nutrients do cycle, returning through decomposers to the soil to be taken up again, but energy does not loop; it flows one way from the Sun and is steadily lost as heat, so it must be resupplied by sunlight

6. Two adjacent plots get identical leaf fall. In plot P the litter layer stays thin all year; in plot Q, after a chemical killed the soil fungi and bacteria, the litter builds into a thick undecayed mat. A visitor concludes 'plot Q's deep litter proves its soil is the more fertile.' The sharpest correction is:

- (A) the visitor is right, because a thicker layer of fallen leaves always means a richer soil beneath it
- (B) the opposite is more likely: in plot P decomposers turn litter into humus and feed the soil, while in plot Q the dead decomposers leave nutrients locked in the unrotted mat, so plot Q's soil is being starved, not enriched
- (C) plot P must have stronger wind that carries its leaves away, so its thin litter says nothing about its soil
- (D) plot Q's trees simply stopped dropping leaves once the mat began to form, which is why its litter looks deep

7. At 2 a.m. a sensor inside a dense forest records the plants taking in oxygen and giving out carbon dioxide. A reporter writes 'so at night a forest behaves like a polluter, proving forests do not really clean the air.' The exact correction is:

- (A) plants do respire and release carbon dioxide at night, but across the whole day the same plants take in far more carbon dioxide and release far more oxygen by photosynthesis, so the net daily effect adds oxygen
- (B) the sensor must be faulty, because plants never take in oxygen under any conditions
- (C) the night reading proves the reporter right, since a forest only releases oxygen on rare days
- (D) photosynthesis runs all night as well, so the night-time carbon dioxide is cancelled out at the very moment it is released

8. A web: leaves to caterpillar to warbler to hawk, and leaves to caterpillar to spider, with the spider also eaten by the warbler. A fungus wipes out nearly all the spiders. The most complete first prediction is:

- (A) with the spider gone, caterpillars lose one predator and tend to rise, while the warbler, having lost its spider prey, leans more heavily on caterpillars
- (B) the warbler starves out at once, because the spider had been its only possible food
- (C) the leaves vanish immediately, because the spiders had been the plants' main protectors against grazing
- (D) the hawk increases straight away, because the spiders had been competing with it for warblers

9. A naturalist labels a chain decaying log to fungus to beetle to shrew and calls the fungus 'the producer, since the chain starts with it.' The precise correction is:

(A) the fungus is a decomposer: it gets its energy by breaking down the dead log rather than by capturing sunlight, so it is not a producer even though it heads this particular chain

(B) the fungus is indeed the producer, because whatever organism a chain starts with must be making its own food

(C) the fungus is a herbivore, because feeding on the plant material in the dead log is the same as grazing a living plant

(D) the fungus is a carnivore, because the beetle that feeds on it treats the fungus as living prey

10. On a hillside an intact forest soaks rain into the soil so a spring below flows all year; the same forest also cools the air and feeds moisture back to the sky. After felling, the slope erodes, the spring floods then dries, and the local air grows hotter and drier. The single underlying loss driving all of these at once is that, with roots and litter gone:

(A) the felled trees had been physically drinking the floodwater and breathing it out as the spring's steady flow

(B) removing the canopy lets sunlight bake the bare soil into a sponge that holds even more water than before

(C) the cut wood gives off steam as it dries, and that lost steam is the sole reason the air becomes drier

(D) rain no longer soaks in slowly but rushes off, so soil is carried away, the spring is no longer steadily recharged, and far less water is transpired and shaded back into the air

11. A pesticide wipes out the soil decomposers across an entire forest, but every plant and animal is left alive at first. A student says 'good, nothing will rot, so the forest will stay clean and grow better.' Reasoning several steps ahead, the most accurate forecast is:

(A) the forest will indeed grow better, because plants make all their nutrients fresh from sunlight and never need the soil

(B) nothing will change, because decomposers only tidy up and play no part in feeding the plants

(C) dead leaves, dung and bodies will pile up unrecycled, the nutrients locked inside will stop returning to the soil, and over time the soil grows poorer so plant growth declines

(D) the plants will at once add new nutrients to the soil themselves, fully replacing the recycling that was lost

12. An undisturbed grassland is watched for sixty years: first more herbs appear, then shrubs, then scattered saplings, then a closed canopy with deep humus and many animals. A boy concludes 'this proves you can grow a complete forest as fast as you like.' The better reading is:

- (A) it shows the opposite: a forest builds up slowly through stages over many years as soil and the community develop, so it cannot be conjured complete on demand
- (B) the boy is right, because the whole sequence shows a layered forest can appear within a single season
- (C) the land was already a finished forest the moment the first extra herbs appeared, and nothing really changed after that
- (D) only deliberate planting in that exact order by people could ever produce such a sequence

13. A tree is pollinated only by one moth and its seeds are dispersed only by one fruit-eating bird. Hunters remove the bird; the moth survives. Over several years the most likely outcome for this tree species is:

- (A) the tree is unaffected, because spreading does not depend on either pollination or seed dispersal
- (B) fruit still sets because the moth still pollinates, but with no disperser few seeds are carried away from the parent, so new saplings rarely start at a distance and the species spreads and renews poorly
- (C) the tree spreads faster, because all its seeds now drop and grow densely right beneath the parent
- (D) the tree dies out at once, because removing the bird also stops the moth from pollinating it

14. A web has a fox that draws on two chains: clover to rabbit to fox, and grass to vole to fox, while an owl also eats the vole. A drought halves the rabbits alone. Tracing two steps, the most reasoned prediction is:

- (A) the fox starves out completely, because the rabbit was its only source of food
- (B) the fox, short of rabbits, leans harder on voles, so voles are pressed more and the owl, now competing with a hungrier fox for the same voles, may get fewer
- (C) the voles increase, because the fox is now too weak from losing rabbits to hunt anything at all
- (D) the clover is wiped out, because halving the rabbits removes the only thing keeping the voles in check

15. A planner compares a forest's products (timber, honey, gum, medicinal herbs) with its services (oxygen, soil holding, water recharge, climate moderation, wildlife shelter). A logger argues the forest's entire worth equals the price of its timber. The strongest counter is:

- (A) the logger is right, because services such as clean air and steady water have no real value to people
- (B) products and services are the same thing, so simply counting the timber already counts everything

(C) a standing forest delivers many life-supporting services at the same time that a stack of cut timber cannot provide, so its real value far exceeds the wood alone

(D) the forest's only genuine value is as shelter for wild animals, with no use to people at all

16. A web: seeds to mouse to owl, seeds to sparrow to owl, and a fox that also eats both the mouse and the sparrow. A disease halves the sparrows only. Tracing two steps, the most reasoned prediction is:

(A) the seeds vanish at once, because the sparrows had been their only consumer

(B) both owl and fox lean more on mice, so mice are hunted harder and tend to fall, and the seeds the mice would have eaten are pressed a little less

(C) the owl and fox both die out, because the sparrow was the sole food of each of them

(D) the mice also halve straight away, because losing the sparrows removes the mice's food supply

17. A logging company burns a large tract of forest whose living trees had been taking carbon dioxide from the air and storing it in wood, leaves and soil humus. The most accurate combined effect on the air is:

(A) oxygen rises, because once the trees are burnt nothing is left in the area to use up the oxygen

(B) nitrogen rises, because burning the wood releases all the nitrogen the trees had stored in it

(C) the carbon stored in the wood and litter is released back as carbon dioxide, and with the trees gone less carbon dioxide is taken in afterwards, so the gas rises on both counts

(D) the air is unchanged, because the carbon the trees held simply disappears the moment they burn

18. A scientist studies an island forest and finds it recycles almost all its own nutrients with very little fresh mineral coming in from outside, and calls it 'self-sustaining.' The best justification is:

(A) its plants manufacture entirely new nutrients out of the sunlight they absorb, so no outside supply is ever needed

(B) rain steadily imports all the nutrients it needs, so the forest never actually has to recycle any

(C) its animals carry fresh nutrients in from the sea each day to replace whatever the plants remove

(D) its decomposers continually return the nutrients in dead leaves, dung and bodies to the soil, so the same nutrients are taken up and reused again and again within the forest

19. A forest pond chain (phytoplankton to small fish to kingfisher) sits beside a bank chain (grass to deer to leopard); one decomposer community on the floor breaks down whatever

dies in either. A spill kills most of the small fish. Reading the shared decomposer link, the most accurate statement is:

- (A) the deer and leopard collapse at once, because they had shared the small fish as a food source
- (B) the kingfisher loses food and tends to decline while ungrazed phytoplankton may build up, and the floor decomposers still recycle the dead from both pond and bank into one shared soil
- (C) the pond and the bank chains are entirely separate, so nothing in the forest connects them
- (D) each chain keeps its own private decomposers, so the pond's dead never reach the bank's soil

20. A forester explains why a centuries-old forest's soil stays rich while a ploughed field beside it grows poorer each year despite the same rainfall. The best combined reason for the forest is:

- (A) the dense canopy keeps out all rain, and the constant dryness by itself is what makes a soil fertile
- (B) the deep shade on the floor by itself enriches the soil, because darkness creates nutrients
- (C) animals churn the soil with their feet, and that trampling is the single source of its richness
- (D) fallen leaves and dead matter are steadily turned into humus by decomposers, while living roots and leaf litter hold that fertile soil in place against rain

21. A tree relies on bats to pollinate its night-opening flowers and on a beetle whose grubs feed only on that tree's rotting fallen fruit, returning the nutrients to the soil near the tree. A pesticide kills the beetles while the bats remain. The most likely consequence is:

- (A) no flowers are pollinated, because killing the beetle also stops the bats from doing their work
- (B) flowers are still pollinated by bats and fruit still forms, but fallen fruit now rots more slowly without the beetle grubs, so nutrient return to the soil near the tree is delayed
- (C) the tree grows faster, because the fallen fruit now stays whole and feeds the roots directly
- (D) the tree is wholly unaffected, since neither pollination nor decay matters to a tree at all

22. A forest manager must choose the action that best keeps a forest's many services running for the long term. The best choice, and the reason, is:

- (A) remove the forest's decomposers so that nothing rots and the forest is kept clean and tidy
- (B) protect the standing forest and replant trees where forest has been lost, so the layered, interlinked community and all its services can persist and recover
- (C) clear the tall canopy trees so that far more sunlight can reach and feed the forest floor
- (D) convert the forest to a single fast-growing crop so the land yields more produce each season

23. A web: grass to grasshopper to frog to snake to eagle. A farmer's poison kills nearly all the snakes. Tracing two steps in each direction, the most complete prediction is:

- (A) the grass vanishes at once, because the snakes had been protecting it from the grasshoppers
- (B) the eagle increases straight away, because the snakes had been competing with it for the frogs
- (C) nothing changes anywhere, because removing a mid-chain animal never affects the rest of the web
- (D) the eagle loses snake prey and tends to decline, while frogs, eaten less, rise and press grasshoppers down, which eases grazing on the grass

24. A child argues 'a forest can never run short of soil nutrients, so we may cart away all the fallen wood and dead leaves for firewood with no harm.' The sharpest flaw in this reasoning is:

- (A) the claim is correct, because plants make all their nutrients fresh from sunlight and never draw on the soil
- (B) removing dead leaves actually helps, because rotting matter only poisons the soil for new plants
- (C) carting off the dead wood and leaves removes the very matter decomposers would have recycled into humus, so over time the soil loses the nutrients it normally gets back
- (D) the soil cannot lose nutrients anyway, since rain delivers a fresh supply of minerals from the sky each day

25. A web: leaves to aphid to ladybird to spider, and leaves to aphid to ant, where the ant farms and protects the aphids from ladybirds. A drought kills almost all the ants. The most reasoned first prediction is:

- (A) aphids rise sharply, because losing the ants removed the aphids' only source of food
- (B) the leaves are stripped bare, because the ants had been the main grazers of the leaves
- (C) the spiders die out, because the ants had been the spiders' chief food all along
- (D) without the ants' protection, ladybirds eat aphids more freely, so aphid numbers tend to fall and the spiders gain more ladybird prey

26. A teacher contrasts a season-long wheat field with a centuries-old forest and asks which single feature most fundamentally makes one a forest and the other not. The best answer is that the forest is:

- (A) simply larger than the field, since the area covered is the only real difference between them
- (B) guaranteed to contain at least one large predator such as a tiger, which a field can never hold
- (C) completely free of insects and animals, whereas a field is always full of them
- (D) a many-species, many-layered community that persists and renews itself over many years, not a single crop grown and cleared in one season

27. On a dense forest floor a fern with broad, thin leaves thrives in deep shade, while a treetop plant with narrow leaves grows in full sun. Asked which adaptation suits the fern, the best answer is that broad, thin leaves let the shade plant:

- (A) store extra water for very dry conditions, since the forest floor faces the harshest, driest sun
- (B) catch and use the small amount of filtered light reaching the dim floor, so it can still make enough food to survive
- (C) make food without any light at all, drawing its energy straight from the humus instead
- (D) shed the flood of excess sunlight that pours onto the floor through a wide-open canopy

28. A naturalist points to a fallen, decaying log busy with fungi, beetles and worms and calls it 'part of the forest's working.' The deepest reason a forest can hold soil, recycle nutrients, refresh air, store water and shelter wildlife all at once is that it is:

- (A) a living, interconnected system whose plants, animals, soil and micro-organisms each support the others, so many services emerge together
- (B) simply a very large stock of timber that happens to provide a few extras while it stands
- (C) a chance collection of unrelated trees growing near one another with no working links
- (D) a man-made arrangement that people must rebuild from scratch each year to keep it working

29. Year after year a forest's plants take nutrients out of the soil to grow, yet the soil does not run dry of them. The best account of how is:

- (A) the plants quietly add brand-new nutrients to the soil out of the sunlight they absorb
- (B) the nutrients never truly leave the soil, since plants only borrow them for a moment and put them straight back

- (C) decomposers keep returning nutrients to the soil from fallen leaves, dung and dead bodies, roughly replacing what the living plants take up
- (D) rain delivers a fresh batch of soil minerals down from the sky every single day

30. A web: leaves to caterpillar to thrush, and leaves to weevil to thrush, with thrush eaten by a hawk. The thrush eats both caterpillars and weevils. A late frost wipes out nearly all the weevils. The most defensible first prediction is:

- (A) the thrush leans more heavily on caterpillars, so caterpillar numbers may be pressed down even though the leaves have lost one of their two grazers
- (B) the thrush starves out completely, because the weevil had been its only food
- (C) the leaves are stripped bare at once, because weevils were their main grazers and caterpillars cannot eat leaves
- (D) the hawk declines straight away, because the weevils had been the hawk's chief food

31. Sunlight filtering through gaps in the canopy reaches the understorey and floor only weakly. Two students argue over whether this dim light matters. The correct view is that the filtered light:

- (A) is just enough for shade-adapted understorey and floor plants to make their own food and survive beneath the canopy
- (B) is harmful to the lower plants and should ideally be blocked out completely
- (C) only warms the floor animals and plays no part in feeding any of the lower plants
- (D) frees the lower plants from needing to take up any water or soil nutrients at all

32. A forest in a wet region returns water vapour to the air through its many leaves and shades the ground from the Sun; removing it makes the area hotter with less reliable rain. The combined mechanism by which the forest had helped bring rain and keep summers mild is:

- (A) the tall trunks physically dragged rain clouds down out of the sky to release their rain
- (B) the trees burned slightly in the heat and the rising smoke seeded the clouds to make rain
- (C) the trees soaked up every bit of the Sun's heat so that the land beneath could never warm at all
- (D) transpiration from its leaves added moisture to the air while its shade and cover kept the ground cooler, together favouring rainfall and milder temperatures

33. A web: clover to beetle to lizard, and clover to grasshopper to lizard, with lizard eaten by a snake. The lizard eats both beetles and grasshoppers. A pesticide wipes out the beetles. Reading two steps ahead, the best prediction is:

- (A) the snake dies out at once, because the beetle had been the base of all its food
- (B) the clover is consumed entirely, because the beetles had been holding the grasshoppers in check

(C) the lizard relies more on grasshoppers, so grasshopper numbers may fall, while the snake, still fed by lizards, is only mildly affected

(D) the lizard population doubles, because losing the beetles leaves more clover for the lizard to eat

34. A bare, ploughed hillside loses far more topsoil in a downpour than the forested slope beside it. The chain of reasons that best explains the difference is that the forest's:

(A) trees had been drinking the rainwater, and now that they are gone the loosened soil simply floats off

(B) canopy had baked the soil dry and hard, and only such hard baked soil ever resists erosion

(C) roots bind the soil and its leaf litter cushions and slows the rain so water soaks in, whereas the bare slope has nothing to grip the soil and rain rushes off carrying it away

(D) shade had cooled the rain, and only cool rain is gentle enough to leave the soil in place

35. A teacher labels the loop soil to roots to plant tissue to animals to dung-and-dead-bodies to decomposers to soil a 'cycle,' and the line grass to deer to tiger a 'chain.' Asked why 'cycle' fits the first but not the second, the best answer is:

(A) the loop's nutrients travel in a perfect circle of exactly equal size every single year, unlike the chain

(B) a chain may have at most three organisms, whereas a cycle is allowed to have more than three

(C) in the loop the same nutrients return to the soil to be taken up again and again, while a food chain runs one way and its energy is lost rather than returned

(D) the loop destroys its nutrients at the end and makes a brand-new set, while the chain reuses one set

36. Two villages sit below identical hills; one hill keeps its forest, the other is cleared. Over a decade the forested village has a steady year-round stream, mild summers and fertile fields, while the cleared one suffers floods, dry summers, harsh heat and thin soil. The best single conclusion is:

(A) the difference is pure chance, because a forest cannot influence streams, climate and soil at the same time

(B) only the flooding differs between them, because climate and soil fertility have no link to forest cover

(C) the cleared hill is actually better off, so the villagers must have confused the two outcomes

(D) the standing forest at once steadied the water flow, moistened and cooled the air, and held the fertile soil, services the cleared hill lost all together

37. A web: nectar plants pollinated by bees, bees eaten by small birds, birds eaten by snakes. A pesticide kills almost all the bees. Tracing both up and down, the most complete

prediction is:

- (A) the birds lose food and tend to decline, the snakes feel the loss one step later, and the nectar plants are pollinated less, so they set fewer seeds and fruit
- (B) only the birds are affected, since pollination and seed-set have nothing to do with the bees
- (C) the nectar plants grow faster, because with no bees taking nectar they keep more energy for seeds
- (D) the snakes increase, because fewer bees means cleaner flowers and so healthier prey for the snakes

38. On a slope an intact forest keeps the water table high so a well below stays full all year. After clearing, the well drops and dries in summer. The best explanation linking forest to groundwater is that, before clearing, the forest:

- (A) let rain soak slowly through litter and root channels into the ground, recharging the water table that fed the well, whereas bare land sheds rain too fast to recharge it
- (B) turned the rainwater into oxygen, which then condensed underground to fill the well
- (C) sealed the soil so tightly that surface water was forced sideways into the well
- (D) sent all of its rainwater straight to the sea, which somehow kept the well topped up

39. A keystone predator such as the tiger keeps deer numbers in check. Poachers remove all the tigers. Following the ripples through several steps, the most likely far-reaching outcome is:

- (A) more deer means more dung, which instantly turns the whole patch into rich new forest
- (B) nothing changes, because the plants and animals in a forest do not really affect one another
- (C) the surplus deer begin hunting and eating the tigers, balancing their own numbers again
- (D) deer multiply, overgraze the young shrubs and saplings, and the forest struggles to renew itself, showing how one removed species disturbs many others

40. A naturalist insists that although the forest floor is the darkest layer, it teems with more crawling and growing life than the brilliant canopy. Resolving this apparent paradox, the best explanation is:

- (A) most light is captured by the layers above, but the floor's piled-up dead matter and damp shelter feed and house vast numbers of decomposers and ground organisms
- (B) it is impossible, since the layer with the least light must always carry the fewest living things
- (C) the floor must really be the brightest layer after all, because it holds the most life
- (D) the floor organisms glow with their own light, so they need none from the Sun above

41. A web: algae to tadpole to fish, and leaf litter to insect larva to fish, with fish eaten by a heron; so the fish draws on a plant chain and a detritus chain. A pollutant kills the algae only. The most reasoned prediction is:

- (A) the heron dies out at once, because the algae had been the base of all of its food
- (B) tadpoles lose food and decline, but the fish can lean on the litter-fed larvae, so the fish and heron are squeezed rather than wiped out
- (C) the fish is unaffected in every way, since algae played no part at all in its food supply
- (D) the leaf litter disappears too, because the algae had been producing the leaf litter

42. A boy claims a burnt forest patch is no real loss because 'we can grow a brand-new full forest there in a few months.' The strongest correction, drawing on the forest being a built-up system, is:

- (A) he is right, because scattering seeds yields a complete grown forest within a single season
- (B) a forest re-establishes slowly over many years as herbs, then shrubs, then trees return and the soil and animal community rebuild, so it cannot reappear complete in months
- (C) the patch will never grow back at all, so the length of time it takes is beside the point
- (D) the forest will regrow within months, but only if a top predator is released into it first

43. A web: grass to zebra to lion, and grass to zebra to hyena, with lions and hyenas also scavenging each other's leftover kills. A long drought destroys nearly all the grass. The most far-reaching effect, and its reason, is:

- (A) only the zebra is hit, because lions and hyenas eat meat and so do not depend on the grass at all
- (B) only the lion is hit, because it alone sits at the very top of the whole web
- (C) every animal in the web suffers, because all of them depend, directly through the zebra or through scavenged kills, on the grass as the producer
- (D) nothing is affected, because lions and hyenas can simply make their own food when the grass fails

44. A drought destroys almost all the producers in a forest, sparing the animals at first, yet within weeks animals at every level are in trouble. The most far-reaching reason is:

- (A) only the plant-eaters are hit, because the meat-eaters get their energy from prey, not from plants
- (B) only the top predator is affected, because it alone sits at the end of every chain
- (C) every animal, herbivore and carnivore alike, ultimately depends on the producers for energy, so losing them cuts the energy base of the whole web
- (D) nothing serious happens, because the animals can make their own food once the plants are gone

45. A child collects 'forest products' for a project and lists honey, gum, medicinal herbs, timber, plastic and petrol. Asked which entries are truly products obtained from the living forest, the correct group is:

- (A) plastic, petrol and timber, since these are the items people most often buy
- (B) honey, gum, medicinal herbs and timber, since all of these come from forest plants and animals
- (C) gum, honey and plastic, since all three are sticky or natural-seeming substances
- (D) petrol, medicines and timber, since each of these is dug or cut from forest land

46. A keystone predator and the soil decomposers of a forest patch are both wiped out in the same year by two separate disasters. Predicting the combined result, the most complete answer is:

- (A) the forest improves, because fewer predators mean more animals and no rotting to spread disease
- (B) the herbivores, unchecked by the predator, overgraze the plants, while dead matter piles up unrecycled and the soil grows poorer, so the forest is hit from two directions at once
- (C) only the predator's loss matters, since the decomposers have no real effect on the forest
- (D) nothing changes, because the two separate losses simply cancel each other out exactly

47. A light study finds the canopy intercepts 80 of every 100 units of incoming light, the understorey absorbs 15 of the 20 that pass it, and the shrub layer takes 4 of the 5 reaching it. The herb layer at the bottom therefore receives, of the original 100 units, about:

- (A) 20 units, because only the canopy removes light and the lower layers add none of their own
- (B) 50 units, because the light is shared roughly equally between the top and the bottom layers
- (C) 0 units, because once the canopy is reached no light of any kind can pass below it
- (D) 1 unit, because each layer above removes most of the light before it can reach the floor

48. A naturalist explains why the high, sunlit branches host fruit bats, hornbills and monkeys while the deeply shaded floor hosts ferns, fungi and beetles. The best account of why the most light and the most decomposer life sit in different layers is:

- (A) the ground actually receives the most light, which is exactly why so many organisms gather there
- (B) the high animals carry all the dead matter up, so the canopy hosts the most decomposers of all

(C) the upper branches intercept most light for the fruit and leaves the high animals feed on, while the dim floor collects the fallen dead matter that feeds the decomposers below

(D) light and dead matter are spread evenly through the forest, so the two layers are really identical

49. A web: leaves to caterpillar to warbler, leaves to caterpillar to shrew (shrews also eat warbler eggs), and warbler and shrew both eaten by an owl. A trap removes all the shrews. The most complete first prediction is:

(A) the owl starves out at once, because the shrew had been its only possible food

(B) warblers, no longer losing eggs to shrews, may increase, caterpillars face one fewer predator, and the owl, missing the shrew, leans more on warblers

(C) the leaves are devoured entirely, because the shrews had been the leaves' main grazers

(D) the caterpillars vanish at once, because they had depended on the shrews for their survival

50. A child says protecting one large forest will save 'only the trees inside it.' Explaining biodiversity, the best correction is that protecting the forest also protects many other species because:

(A) its many layers and varied habitats supply homes and food to a great variety of animals, plants and micro-organisms that all depend on the forest to live

(B) a forest really contains only one kind of organism, so protecting it protects just that one kind

(C) the forest's variety of life actually comes from the soil minerals rather than from the forest itself

(D) the animals could survive equally well on bare land, so saving them does not really need the forest

51. A web: grass to deer to leopard, grass to hare to leopard, and grass to insect to bird to leopard, so the leopard draws on three chains. A disease halves the deer alone. The most reasoned prediction is:

(A) the leopard starves out completely, because the deer had been its only source of food

(B) the grass is wiped out, because the deer had been the only thing keeping the hares in check

(C) the leopard shifts more onto hares and birds, so those prey may be pressed harder, while the grass once eaten by deer is grazed a little less

(D) the hares and birds also halve at once, because they had shared the deer as a food source

52. A forest captures carbon, holds soil, recharges groundwater, refreshes air and shelters wildlife. Asked where the forest stores most of the carbon it removes from the air, the best

answer is:

- (A) in its wood, leaves and soil humus, where the carbon is built into plant tissue and stored as the forest grows
- (B) in the open air packed between the trees, held there tightly by the canopy overhead
- (C) in the fast-flowing streams that run through the forest, dissolved in the running water
- (D) in the fur and feathers of the many different animals that live within the forest

53. Two students name parts of the same forest. Meera points to the leafy roof formed where the branches of many tall trees meet and interlock overhead. Vikram points to the branchy crown of one single tall tree. The terms they are naming, in that order, are:

- (A) the crown of the whole forest, and the canopy of the single tree
- (B) the understorey, and the canopy, since both of these lie below the treetops
- (C) the emergent layer, and the forest floor, which face each other at top and bottom
- (D) the canopy, made by the meeting crowns of many trees, and the crown of one individual tree

54. A web: grass to rabbit to fox, grass to vole to fox and to owl, where fox and owl share the vole. A disease removes all the foxes. Tracing two steps, the most reasoned prediction is:

- (A) the grass vanishes at once, because the foxes had been the only thing keeping the rabbits from grazing it
- (B) the owl declines, because the fox had been the owl's main source of food
- (C) rabbits, losing their only predator, rise sharply, while the owl, no longer competing with the fox for voles, may have more voles to itself
- (D) the voles die out, because the foxes used to supply the voles with their food

55. A scientist watches a patch for a decade before judging it a true forest. The single strongest piece of evidence for that judgement would be:

- (A) many different species of plants, animals and micro-organisms living in interlinked layers and renewing themselves over the whole ten years
- (B) a very large count of tall trees, all of one single species, packed closely together
- (C) the presence somewhere in the patch of at least one large predator such as a tiger
- (D) a thick, even carpet of green grass spread across open ground

56. A pairing quiz lists forest roles against organisms. The pairing that is correct is:

- (A) producer paired with a tiger, because the tiger sits at the very top of the forest
- (B) carnivore paired with grass, because grass feeds the animals beneath it
- (C) herbivore paired with a soil fungus, because the fungus lives on plant remains
- (D) decomposer paired with a mushroom, a fungus that breaks down dead matter into humus

57. A student writes 'forests affect the climate, so cutting them can change only the temperature and nothing else.' Synthesising the chapter, the best correction is that losing a forest tends to change, together:

- (A) only the local temperature, since a forest's sole role in the climate is to provide shade
- (B) air temperature and moisture, rainfall reliability, soil stability, flood and drought patterns, and the air's carbon dioxide level, because the forest influences all of these at once
- (C) only the carbon dioxide level, since gases are the one thing forests actually affect
- (D) nothing measurable at all, because a single forest is far too small to matter to anything

58. In the chain grass to grasshopper to frog to snake to hawk, energy is captured by the grass and passed up with much lost as heat at each step. Which organism's body, on average, traces back to the smallest share of the Sun's original captured energy?

- (A) the grasshopper, because being the smallest-bodied animal it must rely on the least energy
- (B) the frog, because an animal in the middle of a chain always receives the least energy of all
- (C) the grass, because the producer is given the smallest share of energy before any animal feeds
- (D) the hawk, because energy is lost at every transfer and the hawk sits at the top, furthest from the producer

59. A web: grass to grasshopper to frog to snake, and grass to mouse to snake, so the snake eats both the frog and the mouse, and a hawk eats the snake. A late frost wipes out almost all the grasshoppers. Tracing the ripple, the most defensible prediction is:

- (A) the snake starves out completely, because the grasshopper had been the base of all of its food
- (B) the mice die out as well, because losing the grasshoppers removed the mice's only food supply
- (C) frogs lose their grasshopper food and tend to decline, so the snake leans more on mice, and the hawk, still fed by snakes, is only mildly affected
- (D) the hawk increases at once, because the grasshoppers had been competing with it for snakes

60. A teacher draws the nutrient loop soil to roots to plant tissue to grazing animals to dung-and-dead-bodies to decomposers to soil and asks which single cut would most directly stop nutrients ever returning to the soil. The best answer is to remove the:

- (A) decomposers, since without them dung and dead bodies are never broken down into humus and the nutrients stay locked away from the soil
- (B) grazing animals, since they are the only link that turns plant tissue into humus for the soil
- (C) rainfall, since rain is what carries the nutrients out of the dead matter and into the roots
- (D) sunlight, since the whole loop cannot turn at all once the light that drives it is cut off

Solutions — Science (Class 7), Chapter 17:

Forests: Our Lifeline — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	C	21	B	31	A	41	B	51	C
2	A	12	A	22	B	32	D	42	B	52	A
3	B	13	B	23	D	33	C	43	C	53	D
4	C	14	B	24	C	34	C	44	C	54	C
5	D	15	C	25	D	35	C	45	B	55	A
6	B	16	B	26	D	36	D	46	B	56	D
7	A	17	C	27	B	37	A	47	D	57	B
8	A	18	D	28	A	38	A	48	C	58	D
9	A	19	B	29	C	39	D	49	B	59	C
10	D	20	D	30	A	40	A	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) The fern's stated optimum is 10 to 15 units and it is harmed above 50. Only the shrub-layer reading of 12 units lies inside that window. The 100-unit canopy top far exceeds the scorch threshold, so 'more light is always better' fails for a shade-adapted plant. The 3-unit herb floor is below the fern's productive range, so it would make too little food, defeating the idea that darker is automatically safer. The 40-unit understorey is tolerable but below optimum and chosen only by a vague 'middle suits middle' rule rather than by matching the numbers.

2. (A) Each step here loses about ninety percent (60000 to 6000 to 600), so the tiers track the producer base. Tripling the base would push the supportable predator mass towards about three times as much, but because the real fraction passed on varies, the answer is an estimate, not a guaranteed exact 1800. The 'fixed by instinct' option ignores that predator numbers are limited by the energy available below them. The 'cut off from the plant supply' option forgets that a carnivore's energy traces back through its prey to the

producers. The 'gain rather than lose' option inverts the rule that energy is lost as heat at every transfer.

3. (B) From grass to hawk there are four transfers (grass to grasshopper, to frog, to snake, to hawk). Passing on a tenth four times gives a tenth of a tenth of a tenth of a tenth, which is one part in ten thousand. 'About half' ignores the heavy loss at each step. 'Almost all' inverts the rule: the top animal receives the least, not the most. 'None at all' is wrong because energy does flow up the chain, it is just heavily reduced, not stopped.

4. (C) The fox was the rabbit's sole predator but only one of the vole's two (the owl remains). So rabbits are released more fully and tend to climb more steeply than voles, and their grazing presses the shared clover. 'Exactly the same amount' ignores that the vole still has the owl. The clover would be grazed more, not less, so 'clover increases' fails; and the owl eats voles, not clover. The owl and fox were not predator and prey of each other, so the fox's loss does not starve the owl.

5. (D) Nutrients are recycled: dead matter and dung are broken down by decomposers into the soil, where roots take them up again, so that arrow is a loop. Energy is different: it enters as sunlight, is captured by producers, and is lost as heat at every transfer, never returning, so it is a one-way arrow needing constant resupply by the Sun. The option claiming both loop overlooks the heat loss. The option swapping them reverses the truth. The 'nothing moves' option contradicts the very idea of a food chain.

6. (B) Fertility depends on nutrients being released from dead matter into the soil. Plot P's living decomposers do this, keeping litter thin and soil fed; plot Q lost its decomposers, so litter accumulates with its nutrients still locked away, starving the soil. The thick mat is therefore a sign of broken recycling, not richness, refuting 'thicker means richer.' The wind explanation is invented and ignores the stated decomposer kill. Both plots get identical leaf fall, so 'plot Q stopped dropping leaves' contradicts the setup.

7. (A) Forest plants respire day and night, so the night carbon-dioxide reading is real. But by day photosynthesis takes in much more carbon dioxide and releases much more oxygen than respiration uses, so over a full day the forest is a net oxygen supplier. The reporter mistakes one night-time slice for the whole picture. Plants do take in oxygen for respiration, so 'never' is wrong. Photosynthesis needs light and so does not run at night, so it cannot cancel the reading at the moment it occurs.

8. (A) Spiders ate caterpillars and were eaten by warblers, so losing spiders both releases caterpillars (one fewer predator) and removes one of the warbler's foods, pushing the warbler onto caterpillars. The warbler also eats caterpillars, so it does not starve. The spider is a predator of caterpillars, not a protector of leaves, so the leaves are not directly defended by them. The spider was warbler prey, not a competitor of the hawk for warblers, so the hawk does not gain that way.

9. (A) A producer makes its own food from sunlight; the fungus instead lives on the dead log, breaking it down, which makes it a decomposer regardless of its position in this chain. 'Starts the chain therefore producer' confuses position with feeding method. A herbivore eats living plants; the fungus consumes dead, not living, plant matter and is not a plant-eater in that sense. The beetle eating the fungus does not make the fungus a carnivore, that label would describe an animal eating other animals.

10. (D) Roots and leaf litter slow rain so it infiltrates; remove them and rain runs off fast, eroding soil (muddy floods), failing to recharge the spring (it floods then dries), and reducing both transpiration and shading (hotter, drier air). One lost mechanism, infiltration plus tree cover, drives every effect. Trees do not 'drink floodwater and breathe out a spring.' Baking bare soil hardens it and reduces, not increases, water holding. Drying wood releasing steam is trivial and cannot explain erosion, the spring or the heat.

11. (C) Decomposers return nutrients from dead matter to the soil; remove them and dead matter accumulates while its nutrients stay locked away, so the soil is steadily drained and plant growth falls, not rises. Plants need soil minerals (sunlight supplies energy, not minerals), so 'never need the soil' is false. Decomposers do far more than tidy: they are the recycling step, so 'no part in feeding plants' is wrong. Plants take nutrients up, they do not manufacture and add minerals, so they cannot replace the decomposers.

12. (A) The sixty-year sequence of herbs, then shrubs, then trees, with soil and animal life building up, is exactly why a forest cannot be made quickly: it is a slow, staged development. 'A single season' contradicts the decades observed. The forest was not finished at the herb stage; the canopy and humus came much later. The succession happened naturally without people planting, so 'only deliberate planting' is wrong.

13. (B) Pollination (by the moth) continues, so fruit and seeds still form; what is lost is dispersal, so seeds mostly fall under the parent and few new saplings establish elsewhere, weakening the species' spread and renewal. Spread does depend on dispersal, so 'unaffected' is wrong. Seeds crowded under the parent compete and are easily found by pests, so the species does not spread faster. Losing the bird has no effect on the moth, so pollination is not stopped and the tree does not die at once.

14. (B) Losing half the rabbits pushes the fox onto its other prey, the voles; heavier fox predation on voles means the owl, which shares that prey, faces more competition and may take fewer. The fox eats voles too, so it does not starve. A fox short of one food hunts the other harder, so voles are pressed down, not freed to increase. Rabbits eat clover and voles eat grass, so rabbits do not 'keep voles in check', and clover would be grazed less, not wiped out, with fewer rabbits.

15. (C) Cutting the forest for timber gives a one-off product but destroys the continuing services (oxygen, soil, water, climate, habitat) that only a living forest provides, so the standing forest is worth much more than its wood. Clean air and steady water are clearly

valuable to people, so the logger's premise fails. Products and services are different categories, so timber does not 'include' the services. Limiting the value to animal shelter ignores the human-supporting services the chapter stresses.

16. (B) Both the owl and the fox eat mice and sparrows; halving sparrows pushes both predators onto mice, so mice are hunted harder and tend to decline, and with fewer mice the seeds are eaten slightly less. Mice also eat seeds, so sparrows were not the seeds' only consumer. Each predator still has mice, so neither dies out. Mice eat seeds, not sparrows, so losing sparrows does not remove the mice's food.

17. (C) Burning releases the trees' stored carbon as carbon dioxide, and the lost trees no longer remove carbon dioxide by photosynthesis, so the gas rises from both the release and the lost uptake. Oxygen does not rise; burning consumes oxygen, and the claim ignores the carbon released. The chapter's point is about carbon dioxide, not nitrogen. Carbon does not 'disappear' on burning, it becomes carbon dioxide, so the air is not unchanged.

18. (D) Being self-sustaining with little outside input means the same nutrients keep cycling: decomposers return them from dead matter to the soil for roots to reuse, the nutrient cycle of the forest. Plants use sunlight for energy, not to create minerals, so they cannot manufacture nutrients. The premise states little is imported, so 'rain imports all of it' contradicts the setup. Animals do not ferry sea minerals in daily; the recycling is internal.

19. (B) Losing the fish starves the kingfisher (decline) and leaves phytoplankton ungrazed (may build up), while the single decomposer community keeps recycling dead matter from both chains into one soil, which is exactly what links the two chains. The deer and leopard eat grass-based food, not fish, so they do not collapse. The shared decomposers show the chains are connected, refuting 'entirely separate.' Because the decomposer community is shared, the chains do not have private, isolated decomposers.

20. (D) Forest fertility comes from two things working together: decomposers recycling dead matter into humus, and roots plus litter binding the soil so the rich layer is not washed away, both lost when land is ploughed bare. A canopy does not keep out all rain, and dryness does not create fertility. Darkness does not 'create nutrients'; shade alone is not a nutrient source. Trampling is incidental and is not the source of the soil's richness.

21. (B) The bats are unharmed, so pollination and fruiting continue; the loss is the beetle's decomposing role, so fallen fruit rots more slowly and nutrient recycling near the tree is delayed. Killing the beetle does not affect the bats, so pollination is not stopped. Whole, unrotted fruit does not feed the roots; nutrients must be released by decomposers first, so growth does not speed up. Decay matters because it returns nutrients, so the tree is not unaffected.

22. (B) The services (oxygen, soil, water, climate, habitat) flow from the whole living, layered community, so protecting it and restoring lost forest keeps them running. Removing decomposers breaks the nutrient cycle and starves the soil. Clearing the canopy exposes and harms the lower layers and floor life. Converting to one crop replaces the many-species forest with a single-season monoculture, destroying the very community that provides the services.

23. (D) Removing snakes cuts the eagle's food (decline) and releases frogs (rise); more frogs eat more grasshoppers (decline), so grass is grazed less and recovers. Snakes eat frogs, not grasshoppers, so they do not directly protect the grass, and the grass would gain, not vanish. The snake was eagle prey, not a competitor for frogs, so the eagle does not increase. A mid-chain removal clearly ripples both ways, so 'nothing changes' is wrong.

24. (C) The soil stays rich only because dead leaves and wood are recycled into humus by decomposers; removing that dead matter cuts off the nutrient return, so the soil is gradually drained. Plants need soil minerals, so sunlight does not free them from the soil. Rotting matter enriches rather than poisons the soil, so removing it does not help. Rain does not deliver a daily mineral supply from the sky, so the soil can indeed lose nutrients.

25. (D) The ants shielded aphids from ladybirds; remove the ants and ladybirds prey on aphids freely, so aphids fall and ladybirds (now better fed) give the spiders more prey. The ants protected aphids, they were not the aphids' food, so aphids do not rise from losing a food source. The ants did not graze leaves, so the leaves are not stripped. The ants were not spider food, so the spiders do not starve.

26. (D) A forest's defining nature is being a diverse, layered, self-renewing community lasting many years, unlike a monoculture grown and harvested in a single season. Size is not the essential difference; a small patch can be a forest. A forest need not contain a particular large predator to count as one. A forest is full of animals and insects, so 'free of animals' is the reverse of the truth.

27. (B) Broad, thin leaves present a large surface to gather the little filtered light at the floor, letting the shade fern make enough food in dim conditions. The floor is shaded and damp, not the harshest dry sun, so water storage is not the point. Plants cannot photosynthesise without light or draw energy from humus, so 'no light at all' is wrong. The canopy blocks most light, so the floor is dim, not flooded with sun to be shed.

28. (A) The many services arise because the forest is an interlinked community in which each part supports the others, exactly what the decaying log illustrates. Calling it 'a stock of timber' misses the living interactions. The trees are not unrelated; they are linked through soil, food webs and shared decomposers. A forest renews itself naturally and is not rebuilt by people each year.

29. (C) The balance is kept by recycling: decomposers break down dead matter and dung, returning nutrients to the soil to roughly match what plants remove. Plants use sunlight for energy, not to manufacture minerals, so they do not add new nutrients from light. Plants do take nutrients up into their tissue, so 'never leave the soil, only borrowed for a moment' is wrong. Rain does not deliver daily mineral supplies from the sky.

30. (A) Losing weevils pushes the thrush onto caterpillars, pressing caterpillar numbers down; the leaves lose one grazer (the weevil) but gain pressure on the other (more caterpillars eaten), a mixed effect. The thrush still eats caterpillars, so it does not starve. Caterpillars do eat leaves, so the leaves are not left to a single grazer and are not stripped bare. The hawk eats thrushes, not weevils, so losing weevils does not directly cut the hawk's food.

31. (A) Shade-adapted plants are tuned to the weak filtered light and use it to photosynthesise enough to live, so the dim light matters. It is not harmful and should not be blocked; without it the lower plants could not make food. It does feed the plants, not merely warm animals. Photosynthesis uses light but plants still need water and soil nutrients, so the light does not free them from those needs.

32. (D) Forests influence local climate through two real processes: transpiration adds moisture that favours rainfall, and shade plus cover keep the ground cooler, so removing them brings heat and less reliable rain. Trunks do not drag clouds down. Living trees do not burn and seed clouds with smoke. Shade reduces heating but does not absorb every bit of heat so the land 'never warms', which is an overstatement.

33. (C) Losing beetles pushes the lizard onto grasshoppers, pressing grasshopper numbers; the lizard still has a food source, so the snake above it is only mildly affected. The snake does not die out, since the lizard persists. Beetles and grasshoppers both eat clover but beetles do not 'hold grasshoppers in check', so the clover is not consumed entirely. The lizard is a carnivore and does not eat clover, so it does not double from extra clover.

34. (C) The forest resists erosion because roots hold the soil and litter slows the rain so it infiltrates rather than running off; the bare slope lacks both, so rain sheds fast and carries soil away. Trees do not stop soil 'floating off' by drinking rain. A canopy does not bake soil hard; it shelters it. Cooling the rain is not how forests prevent erosion, and 'only cool rain is gentle' is invented.

35. (C) 'Cycle' fits because the same nutrients are returned to the soil and reused repeatedly, whereas the food chain carries energy one way and loses it as heat, so it does not loop. The cycle need not be a perfect equal-sized circle each year, that exactness is invented. The difference is the return path, not a limit on the number of organisms. The loop reuses the same nutrients, it does not destroy them and make new ones, that option reverses the truth.

36. (D) All the contrasts (steady stream, mild summers, fertile soil versus floods, dry summers, heat, thin soil) trace to the forest's combined services: water regulation, climate moderation and soil holding, lost together when cleared. These are real, linked forest services, so it is not chance. Climate and soil fertility are linked to forest cover, so it is not 'only flooding.' The cleared hill is plainly worse off, so the villagers did not confuse the outcomes.

37. (A) Removing bees works both ways: birds lose prey (decline), snakes lose prey one step later, and the plants lose their pollinator so they set fewer seeds and fruit. Pollination depends on the bees, so 'nothing to do with the bees' is wrong. Plants pollinated less set fewer seeds, not more, so they do not grow faster. Fewer bees means less food up the chain, not 'cleaner flowers and healthier prey', so the snakes do not increase.

38. (A) Forest litter and root channels let rain infiltrate slowly, recharging the groundwater that feeds the well; bare land sheds rain fast, so recharge fails and the well dries. Rain is not turned into oxygen, and oxygen does not condense into well water. The forest loosens, not seals, the soil; sealing would prevent infiltration. Sending rain to the sea would empty, not top up, the well, so that option contradicts itself.

39. (D) Without the tiger, deer multiply unchecked and overgraze young plants, so the forest cannot renew itself, a clear example of one species' removal rippling widely. Extra dung does not 'instantly' create forest; the overgrazing harm dominates. Forest species are interdependent, so 'nothing changes' is wrong. Deer are herbivores and cannot hunt tigers, so they cannot balance their own numbers that way.

40. (A) Light feeds producers, but the floor's abundance of dead matter and damp shelter supports huge numbers of decomposers and ground feeders, so the darkest layer can hold the most life. 'Least light means fewest organisms' assumes all life depends directly on light, which decomposers do not. The floor is not actually the brightest layer; the canopy intercepts the light. Floor organisms do not glow to make their own food; they live on dead matter.

41. (B) Killing algae starves the tadpoles (decline), but the fish still has the litter-fed larvae, so the fish and the heron are squeezed, not eliminated. The heron does not die out at once, since the litter chain still supplies the fish. The fish is partly affected (its tadpole supply drops), so 'unaffected in every way' is wrong. Leaf litter comes from fallen leaves, not from algae, so it does not disappear with the algae.

42. (B) A forest is a built-up, layered community whose soil and life develop in stages over many years, so it cannot be regrown complete in months. Scattering seeds does not yield a grown forest in one season. A burnt patch can recover slowly, so 'never grows back' is wrong. Regrowth follows succession, not the introduction of a predator, so releasing a top predator does not make it regrow in months.

43. (C) Grass is the producer that feeds the zebra, which feeds the lion and hyena, which also scavenge those kills, so destroying the grass cuts the energy base for the whole web and every animal suffers. The carnivores depend on the grass through their prey, so 'only the zebra' is wrong. The lion is not the only one hit; the hyena and zebra suffer too. Animals cannot make their own food, so 'nothing is affected' is impossible.

44. (C) All animal energy traces back to producers, so destroying them removes the base that feeds herbivores and, through them, carnivores, so every level suffers. Meat-eaters depend on prey that depend on plants, so 'only plant-eaters' is wrong. Every level, not just the top predator, loses its energy source. Animals cannot make their own food, so 'nothing serious happens' is false.

45. (B) Honey, gum, medicinal herbs and timber all come directly from forest plants and animals and are genuine forest products. Plastic and petrol are manufactured or fossil products, not obtained from a living forest, so any group containing them is wrong. Plastic is not a natural forest substance, so the 'sticky or natural-seeming' grouping fails. Petrol is not dug from forest land as a forest product, so that group is also wrong.

46. (B) Two key roles are lost together: without the predator, herbivores multiply and overgraze; without decomposers, dead matter accumulates and the soil is no longer fed, so the forest is harmed on both fronts. More herbivores overgrazing and a starved soil is harm, not improvement. Decomposers are essential to the nutrient cycle, so their loss does matter. The two losses both damage the forest, so they do not cancel out.

47. (D) Of 100 units, 80 are taken by the canopy, leaving 20; the understorey takes 15 of those, leaving 5; the shrub layer takes 4, leaving 1 for the herb layer. So about 1 unit reaches the floor. '20 units' ignores the understorey and shrub absorption. '50 units' invents an equal split that the figures contradict. '0 units' is wrong because some light does filter through, just very little, which is why shade plants survive.

48. (C) Light is captured highest, where leaves and fruit and the animals feeding on them live; dead matter falls and collects on the dim floor, feeding the decomposers there, so the two resources, and the life depending on each, sit in different layers. The ground is the darkest, not the brightest, layer. Dead matter falls down to the floor, it is not carried up to the canopy. Light and dead matter are not spread evenly, so the layers are not identical.

49. (B) Removing shrews helps warblers (no egg loss) and caterpillars (one fewer predator), and pushes the owl onto warblers; that is the fullest first picture. The owl also eats warblers, so it does not starve. Shrews ate caterpillars and warbler eggs, not leaves, so the leaves are not stripped. Shrews preyed on caterpillars, so caterpillars do not depend on shrews for survival and do not vanish.

50. (A) A forest's layered, varied habitats house and feed a great diversity of species, so protecting the forest protects all of them, not just the trees. A forest contains many kinds

of organisms, not one, so that option is false. The variety of life depends on the forest's habitats, not merely on soil minerals. Forest species depend on the forest, so they could not survive equally well on bare land.

51. (C) With deer halved, the leopard turns to its other prey (hares and birds), pressing them harder, while the deer's share of grazing eases. The leopard has three prey, so it does not starve. Deer do not control hares (both eat grass), so halving deer does not unleash hares to wipe out the grass; grass is grazed slightly less. Hares and birds eat grass and insects, not deer, so they do not halve from losing the deer.

52. (A) Trees take in carbon dioxide and build the carbon into wood, leaves and, via litter, soil humus, which is where the bulk of forest carbon is stored. It is not held loose in the air between trees, that would not be 'stored.' Streams carry little of the forest's carbon compared with the biomass and soil. Animal fur and feathers hold only a tiny fraction, not most of it.

53. (D) The interlocking leafy roof of many trees is the canopy; the top of one single tree is its crown, so Meera names the canopy and Vikram the crown. The first option swaps the two terms. The understory lies below the canopy, not at the top, so it does not name either. The forest floor is the bottom layer, not a single tree's top, so that pairing is wrong.

54. (C) Removing foxes releases the rabbits (their only predator), so rabbits rise, and frees voles from one of two predators, so the owl faces less competition and may take more voles. The grass would be grazed more by extra rabbits, not 'vanish at once' because foxes protected it. The fox was not the owl's prey, so the owl does not decline. Foxes ate voles, they did not feed them, so the voles do not starve.

55. (A) A forest is defined by a diverse, layered, self-renewing community over time, so observing that is the strongest evidence. A dense stand of one species is a plantation, not the diverse community that defines a forest. One large predator is neither necessary nor sufficient to make a patch a forest. A grass carpet describes grassland, not a layered forest.

56. (D) A mushroom is a fungus that breaks down dead matter into humus, the very definition of a decomposer, so that pairing is correct. A tiger is a carnivore, not a producer; producers are green plants. Grass is a producer, not a carnivore. A soil fungus that lives on dead plant remains is a decomposer, not a herbivore, which eats living plants.

57. (B) A forest influences temperature, moisture and rainfall, soil stability, flood and drought patterns and carbon dioxide together, so losing it changes all of these, not just temperature. Shade is only one of many roles, so 'only temperature' is too narrow. Forests affect water, soil and climate as well as gases, so 'only carbon dioxide' is too narrow. Forests have measurable effects, so 'nothing at all' is wrong.

58. (D) Energy falls at each transfer, so the organism furthest from the producer, the top hawk, traces back to the smallest share. Body size does not set the energy share, so the

grasshopper is not chosen for being small. Middle animals do not always get the least; the top gets the least, so the frog is wrong. The grass is the producer with the most captured energy, not the least.

59. (C) Killing grasshoppers starves the frogs (decline), pushing the snake onto its other prey, the mice; the hawk still has snakes, so it is only mildly affected. The snake also eats mice, so it does not starve. Mice eat grass, not grasshoppers, so losing grasshoppers does not remove the mice's food. Grasshoppers are low in the web and do not compete with the hawk for snakes, so the hawk does not increase.

60. (A) Decomposers are the step that breaks dung and dead bodies into humus, returning nutrients to the soil; cut them and the nutrients stay locked in undecayed matter, so this break most directly stops the return. Grazing animals pass nutrients on but do not make humus, so removing them is not the key break. Rain does not carry nutrients out of dead matter into roots; decomposers release them into the soil first. Sunlight powers plant growth but is not the link that returns nutrients to the soil, so cutting it does not directly stop the return.